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LIQUIDITY CHAIN

LQC Whitepaper

Modular Global Liquidity Infrastructure Connecting Fragmented Web3 Liquidity

DESIGN PRINCIPLE

Liquidity First - Utility Second - Revenue Third - Scale Last

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Document Notice

This document is a technical and business working draft. It distinguishes the confirmed direction of LQC from current design criteria, proposals under review, and areas requiring further research. It does not guarantee a launch, exchange listing, partnership, revenue, token price, or investment return.

Status

Meaning

Confirmed direction

A project objective, principle, or disclosure standard adopted by LQC

Design criterion

A current engineering baseline that may change after audit, simulation, or testing

Under review

A proposed value or mechanism awaiting a final decision

Future research

A topic requiring separate technical, legal, and economic validation

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1. Executive Summary

LQC is not intended to be another standalone exchange. It is designed as an infrastructure layer for discovering, comparing, connecting, and safely using liquidity across multiple markets.

Digital-asset liquidity is fragmented among decentralized exchanges, blockchains, bridges, and individual pools. As a result, the same asset can trade at different prices and costs across venues, while larger orders face higher price impact and execution risk. LQC aims to connect validated liquidity sources through a unified routing layer and progressively extend that infrastructure into collateralized lending, cross-chain movement, staking, and enterprise liquidity services.

Pillar

Role

Current classification

Multi-DEX Liquidity

Register and connect liquidity from multiple DEXs and pools

Phased implementation

Smart Router

Compare price, fees, gas, slippage, and price impact

Core product

Lending

Approved stable-asset loans collateralized by LQC

Design criterion

Oracle & Risk

External feeds, DEX TWAP, caps, and Health Factor controls

Essential safety layer

Cross-chain

Message validation and chain-level supply reconciliation

Phased expansion

Treasury

Accounting for fees, reserves, liquidity, and ecosystem budgets

Design criterion

LQC must be evaluated by measurable price improvement, liquidity depth, execution success, lending health, security audits, and on-chain transparency - not by token narrative alone.

2. Market Problem and the LQC Approach

Liquidity fragmentation increases transaction cost, reduces capital efficiency, and expands systemic risk.

Market problem

Effect

LQC direction

Fragmented liquidity

Price differences, slippage, and shallow markets

Connect multiple pools through one routing layer

Isolated chains

Bridge complexity and added security exposure

Simplify access through validated cross-chain routes

Low capital efficiency

Assets must often be sold to obtain liquidity

Enable conservative collateralized borrowing

Incentive dependence

High-emission APY becomes recurring sell pressure

Base rewards on realized revenue and approved budgets

Opaque revenue

Sustainability depends on token price

Build Treasury funding from service usage

2.1 Design Principles

A DEX or pool is not routable until usable liquidity is verified.

Test, audit, liquidity, and legal gates take priority over a target launch date.

No single administrator key should control supply, Treasury, Oracle, Bridge, and upgrades.

Features and parameters must be disclosed as confirmed, designed, under review, or research.

3. Protocol Architecture

LQC separates modules to limit failure domains and clarify audit, upgrade, and emergency responsibilities.

Layer

Example contracts

Primary responsibility

Token

LQCToken.sol

Fixed cap, transfer, burn, bridge mint controls

Liquidity

LiquidityVault.sol, PoolManager.sol

Asset accounting and pool registry

Router

Router.sol

Quote, optimize, split, and execute routes

Lending

CollateralVault.sol, LendingPool.sol

Collateral, borrowing, repayment, and withdrawal

Risk

BorrowManager.sol, LiquidationEngine.sol

Borrow limits, Health Factor, and liquidation

Oracle

OracleManager.sol

Validated price and oracle state

Revenue

FeeManager.sol, Treasury.sol

Fee collection, allocation, reserve, and burn

Cross-chain

BridgeManager.sol

Message verification, nonce consumption, and supply conservation

Control

AccessManager, EmergencyController

Role separation, timelock, and module-level pause

3.1 Core Invariants

A swap must not execute below the user's minimum output or after its deadline.

Cross-chain movement must not create unexplained LQC supply.

Oracle anomalies pause new risk while preserving safe repayment wherever possible.

Protocol revenue and Treasury movements must be traceable through events and public addresses.

4. Multi-DEX Liquidity Layer

The objective is not to create as many pools as possible, but to secure liquidity that can reliably absorb trades and liquidations. Candidate chains, DEXs, and pools are evaluated by depth, execution stability, oracle quality, security history, user base, and operating cost. A platform is not described as an official partner unless the relationship is independently confirmed.

Selection criterion

Example measurement

Market depth

Price impact by order size, spread, daily volume

Pool health

LP concentration, withdrawal risk, token restrictions

Technical stability

Router compatibility, failure rate, audit and incident history

Operability

Gas cost, RPC reliability, monitoring and emergency response

Price validation

External feeds and sufficiently deep TWAP availability

Usable execution and liquidation capacity matter more than a headline liquidity number.

5. LQC Smart Router

The Smart Router optimizes both the user's net received amount and the reliability of execution.

$$\text{Net Output} = \text{Expected Output} - \text{Pool Fee} - \text{Protocol Fee} - \text{Gas Cost} - \text{Risk-adjusted Cost}$$

Step

Processing

Quote

Collect quotes from registered DEXs and pools

Validate

Check pool state, token rules, price impact, and expiry

Optimize

Rank candidates by net output and execution risk

Protect

Apply minOut, slippage, deadline, and allowlist rules

Execute

Execute atomically through one route or a split route

Account

Record fees, events, and Treasury attribution

The circular and elliptical path visualization used on the LQC website abstracts the routing graph. Nodes represent chains, DEXs, and pools; curves represent possible swap or bridge connections. The Router does not simply choose the shortest line. It chooses the lowest weighted cost based on price, depth, gas, time, security, and failure probability.

Split orders when a single pool would create excessive price impact.

Enforce minimum output, allowed slippage, deadline, and approved paths.

Evaluate quote lifetime and private execution options as MEV defenses.

Expose route, expected fees, and price-improvement rationale through the SDK/API.

6. Collateralized Lending Protocol

Borrow limits are based on realistically liquidatable collateral value and position health, not headline market value alone.

Parameter

Current design criterion

Required validation

Borrow asset

Approved stable assets such as USDT/USDC

Reserve, repayment liquidity, and depeg risk

Initial LTV

45% technical baseline

Volatility, depth, legal review, and simulation

Additional borrowing

OFF by default; explicit opt-in

Recheck HF, oracle state, and maximum debt

Additional limit

Up to 10% of collateral-value increase

Also subject to Borrow Cap and market depth

Position health

$HF = \text{collateral value} \times 60\% / \text{total debt}$

Liquidatable at $HF \leq 1$

Liquidation

Partial liquidation first; 50% Close Factor baseline

5% base penalty; governance ceiling 10%

Bad-debt defense

Debt Ceiling, Borrow Cap, and Safety Reserve

Pause new borrowing when bad debt occurs

For 10,000 USDT of collateral and a 45% maximum LTV, the initial maximum loan is 4,500 USDT.

At a 60% liquidation threshold, opening HF is 1.333. All values require pre-launch risk simulation and audit.

6.1 Lending and Repayment Fees

Fees may be paid directly in LQC or paid in a stable asset that is subsequently used to purchase LQC. Initial net-fee allocation is designed as Burn 20%, Safety Reserve 50%, and Ecosystem/Development 30%. After the disclosed reserve target is achieved, the allocation may

transition to 50%, 30%, and 20%, respectively. Burn affects supply but does not guarantee price appreciation.

7. Oracle, Liquidation, and Risk Management

Risk is assessed from collateral, debt, liquidity, and price-feed health together.

Risk

Scenario

Mandatory control

Price collapse

Collateral falls faster than debt

Conservative LTV, buffer, asset caps

Oracle manipulation/failure

Incorrect lending or liquidation price

External feed + TWAP, deviation and freshness limits

Insufficient liquidity

Market cannot absorb liquidation

Depth-based limits, partial liquidation, reserve

Smart-contract failure

Theft, freeze, or accounting error

Audit, bug bounty, phased TVL cap

Stablecoin depeg

Debt and reserve values diverge

Asset-level caps and pause rules

Administrator risk

Key compromise or discretionary action

Multisig, timelock, role separation

Oracle priority is: validated external oracle; sufficiently liquid DEX TWAP; deviation, staleness, sequencer, and chain-state checks. An anomaly pauses new and additional borrowing while prioritizing repayment. A single shallow LQC pool cannot be the sole lending price source.

8. Cross-chain Liquidity

Message reliability and supply reconciliation take priority over expansion speed. Ethereum, BNB Chain, and Base are candidate environments, not confirmed deployments. The reference architecture locks canonical tokens in a source-chain vault and mints representations on the destination chain. Return movement burns the representation before unlocking the canonical asset.

Each message includes sourceChainId, nonce, amount, recipient, and deadline. A consumedNonce mechanism permits one execution only. Providers and target chains are finalized only after audit and liquidity

validation.

Approach

Benefit

Primary risk

Bridge integration

Faster market expansion

Bridge exploit and wrapped-asset trust

Messaging protocol

Cross-chain application composability

Validator and relayer risk

Chain-native liquidity

Local ecosystem access

Re-fragmentation and operational complexity

Intent-based routing

Simpler user experience

Solver concentration and settlement risk

Across native, locked, minted, and burned balances on every chain, no unexplained LQC may exist.

9. LQC Token Utility

Long-term utility must come from protocol usage rather than price expectations.

Use

Activation condition

Liquidity asset

Approved DEX pools and ecosystem liquidity

Collateral

Lending, Oracle, and risk parameters approved

Lending/repayment fee

Fee policy and user-protection review completed

Gas benefit

Discount or sponsorship defined in Router design

Staking

Funded by realized revenue or an approved emissions budget

Governance

Legal, security, and economic model approved

Ecosystem access

Available only for launched, contracted, or approved products

Mainnet gas/validator

Only if an independent LQC mainnet is implemented

LQC rejects an unlimited-emission model marketed through high APY. Reward funding, issuance limits, duration, stopping conditions, and dilution must be disclosed in advance.

10. Revenue, Treasury, and Burn

Cash flow from actual service usage is intended to fund operations and security.

Potential revenue

Trigger

Attribution principle

Router/Swap fee

Completed trade

FeeManager to Treasury

Lending interest/fee

Active lending service

Reserve, operations, Safety Module

Cross-chain fee

Transfer or messaging

Security, Treasury, ecosystem

Liquidity-as-a-Service

Executed B2B agreement

Contract-specific policy

Usage growth -> fee growth -> Treasury/Reserve accumulation -> reinvestment in security, liquidity, and product -> improved service quality -> reuse

Buyback, burn, or distribution may be executed only from realized net revenue, subject to legal review, governance approval, and disclosed limits. LQC excludes circular structures that sell newly issued tokens to fund buybacks while operating at a loss.

11. Tokenomics

Maximum supply is fixed at **1,000,000,000 LQC**. Planned initial circulation at TGE is **150,000,000 LQC (15%)**.

Allocation

Share / amount

TGE / cliff

Vesting or execution

Future Ecosystem Rewards

35% / 350M

0 at TGE

7+ years; maximum 50M annually

Community Initial

20% / 200M

80M at TGE

Remaining 120M activity-based over 24 months

Team & Core Contributors

20% / 200M

0 at TGE; 12-month cliff

Monthly over the following 36 months

Protocol Treasury

10% / 100M

10M at TGE

Remaining 90M under a five-year budget framework

Liquidity & Market Making

10% / 100M

50M at TGE

Remaining 50M linked to exchange and pool growth

Grants & Strategic Ecosystem

5% / 50M

10M at TGE

Remaining 40M milestone-based

The contract cap must equal maximum supply. Arbitrary additional minting is prohibited. Wallet addresses, monthly unlocks, and circulation calculations will be disclosed and reconciled before deployment and listing applications.

12. Security, Governance, and Transparency

Technical safeguards and operational authority disclosures are product requirements.

Domain

Control

Treasury

Multisignature, spending policy, public reporting

Oracle changes

Role separation, timelock, deviation limits

Token supply

Fixed cap, vesting, public events

Bridge

Chain-level rate limits and supply accounting

Upgrade

Timelock, advance notice, external review

Emergency

Module-level pause, recovery conditions, post-incident report

Pre-launch gates include an independent contract audit, economic/oracle/liquidation/bridge threat modeling, bug bounty, phased TVL and loan caps, key-recovery drills, publication of official contract and Treasury addresses, and a documented incident-response process.

13. Phased Roadmap

Passing conditions take priority over dates.

Phase

Main output

Gate

Phase 0 - Design

Whitepaper, tokenomics, threat model, legal analysis

Public core parameters and authority model

Phase 1 - Foundation

Token policy, Vault, basic Router, Treasury

Initial liquidity, controls, and tests

Phase 2 - Router MVP

Multi-DEX, split routing, SDK/API

Audit, caps, price improvement, execution rate

Phase 3 - Lending

Collateral, Oracle, HF, liquidation

Liquidation liquidity, simulation, legal review

Phase 4 - Cross-chain

Bridge control and supply accounting

Audit, rate limits, recovery exercise

Phase 5 - Alliance

Wallet, DeFi, MM, and enterprise APIs

Product KPIs and integration evidence

Future Research

AI optimization, RWA, independent mainnet

Separate technical, legal, and economic validation

14. KPIs and Public Reporting

Success is measured through liquidity, product, risk, revenue, and security data - not token price.

Category

Core metrics

Liquidity

TVL, pool depth, spread, LP concentration, uptime

Trading

Router volume, executions, price improvement, failure rate, realized slippage

Lending

Outstanding loans, utilization, average HF, liquidations, bad debt

Revenue

Gross fees, net revenue, operating cost, Treasury/Reserve movement

Security

Audit issues, incidents, pauses, bug-bounty resolution

Token

Actual circulation, unlocks, holder concentration, LP status

Adoption

Active wallets, repeat usage, verified integrations, API usage

LQC intends to publish KPIs, circulation, vesting, Treasury movements, security issues, and parameter changes monthly or at another pre-announced interval.

15. Team, Entity, and Ecosystem Relationships

Only verifiable entities and roles are disclosed. Discussion, technical integration, contract, and investment must be clearly distinguished.

Item

Current disclosure standard

Entity

MMXlabs&LQC LLC, Wyoming, United States; formed April 30, 2026

Team structure

4 engineers and 2 advisors; founder, officers, and UBO provided separately for exchange KYC

Public repository

github.com/LQC-labs/LQC-Liquidity-chain

Official channels

Website, X, Telegram, and support email must match across submissions

Relationship labeling

No official partner/listing claim before supporting evidence exists

Conflict disclosure

Separate team, Treasury, market-making, early-investor, and advisor wallets

16. Risk Factors and Legal Notice

LQC is a high-risk digital-asset project exposed to smart-contract, market, liquidity, operational, and regulatory risk.

Risk

Examples

Mitigation direction

Technical

Contract, Oracle, Bridge, key failure or attack

Audit, role separation, pause, caps

Market/liquidity

Crash, shallow pool, manipulation, failed liquidation

Conservative LTV, depth limits, reserve

Token economics

Concentration, unlocks, dilution, insufficient demand

Vesting, circulation, wallet disclosure

Operations

Delay, concentrated authority, conflicts

Multisig, timelock, periodic reports

Relationships

Termination or overstated cooperation

Evidence of contract/integration scope

Regulation

Token, lending, staking, and revenue rules

Jurisdictional review, AML/KYC, restrictions

This working draft is provided for information only. It is not investment, legal, or tax advice; a securities filing; an offer; or a guarantee of return. LQC price, listing, revenue, market share, launch, and roadmap delivery are not guaranteed.

17. Conclusion

LQC should be judged as a real product that safely connects fragmented liquidity. MMX Labs & LQC prioritizes a Router-centered MVP, verifiable liquidity, conservative lending risk management, transparent Treasury operations, and phased security validation. Multi-DEX, lending, cross-chain, and ecosystem expansion are successive parts of one liquidity layer.

One-sentence definition: LQC aims to be modular global liquidity infrastructure that discovers, compares, and connects fragmented digital-asset liquidity and uses it safely across trading, lending, and cross-chain services.

Appendix A. Key Terms

Term

Definition

DEX

A decentralized exchange executing trades through smart contracts

Pool

On-chain liquidity available for a specific asset pair

Router

A module that compares routes and costs and executes trades

TVL

Total value of assets deposited in a protocol

LTV

Maximum loan relative to collateral value

Health Factor

Ratio of adjusted collateral value to debt

TWAP

Time-weighted average market price

Slippage

Difference between expected and executed price

MEV

Value extracted through transaction ordering or related techniques

Timelock

Mandatory delay before an administrative change takes effect

Appendix B. Decisions Required Before Final Publication

Publish the wallet-level TGE calculation, addresses, and evidence for 150M initial circulation.

Finalize allocation and vesting for team, investors, ecosystem, liquidity, and Treasury.

Finalize mint authority, supply cap, administrator roles, and multisig structure.

Finalize initial chains, DEXs, pools, and technical integration scope.

Finalize LTV, liquidation threshold, Close Factor, bonus, and Debt Ceiling.

Finalize fee asset, conversion method, and evidence for Burn/Reserve/Ecosystem execution.

Finalize oracle provider, TWAP, deviation, and freshness thresholds.

Finalize legal opinion, auditor, bug bounty, incident owner, team credentials, and relationship evidence.

18. Smart Contract Functional Specification I - Core Architecture

The implementation separates Liquidity, Lending, Cross-chain, and shared services.

Layer

Module

Main contracts

Responsibility

Liquidity

Router, Pool, Vault

Router.sol, PoolManager.sol, LiquidityVault.sol

Quotes, routes, registry, accounting

Lending

Collateral, Borrow

CollateralVault.sol, LendingPool.sol, BorrowManager.sol

Collateral, supply, limits, debt

Risk

Rate, Liquidation

InterestRateModel.sol, LiquidationEngine.sol, OracleManager.sol

Rate, validation, HF, liquidation

Cross-chain

Bridge

BridgeManager.sol

Message validation, limits, global supply conservation

Shared/Control

Value, Authority

LQCToken, FeeManager, Treasury, Access, Governance, Emergency

Token, fees, finance, authority, pause

Implementation sequence: Modular Architecture -> Role Separation -> Oracle Protection -> Risk Control -> Emergency Protection -> External Audit.

19. Functional Specification

II - Token, Liquidity, and Routing

Contract

Responsibility

Main functions

Protection

LQCToken.sol

Ecosystem token

transfer, approve, burn

Fixed cap; minimal mint authority

LiquidityVault.sol

Protocol asset accounting

deposit, withdraw, balance

Approved-contract access only

Router.sol

Optimal DEX route and execution

getQuote, findBestRoute, executeSwap, splitSwap

Slippage, gas, fee, and deadline validation

PoolManager.sol

Pool registry and status

add/remove, enable/disable, liquidity query

Governance/multisig approval and events

Split execution occurs only when net price improvement exceeds added gas and failure risk. Pool status changes and exclusions are recorded through on-chain events and operating reports.

20. Functional Specification III - Lending, Oracle, and Liquidation

Contract

Main functions

Core logic

Pre-deployment decision

CollateralVault.sol

deposit/withdrawCollateral, balance, value

LQC custody; HF check after withdrawal

Assets and withdrawal conditions

LendingPool.sol

supply, withdraw, borrow, repay

Stable-asset supply and loan liquidity

Assets, reserve, caps

BorrowManager.sol

borrowLimit, debt, validateBorrow, HF

Limit = collateral value x LTV

Final LTV after simulation

InterestRateModel.sol

utilization, borrow/supply rate

Utilization-based variable rate

Curve, kink, maximum utilization

Liquidation + Oracle

check/liquidate, price/TWAP/deviation

HF liquidation with validated pricing

HF, Close Factor, staleness, deviation

A single LQC pool price cannot determine lending or liquidation. Oracle failure restricts new borrowing and preserves repayment where safe.

21. Functional Specification IV - Fee, Treasury, Bridge, and Control

Contract

Function

Control principle

FeeManager.sol

Aggregate Router, Lending, Bridge, and service fees

Record policy, transfer, and burn events

Treasury.sol

Revenue, reserve, incentives, optional buyback/burn

No personal wallet; multisig + timelock

BridgeManager.sol

Movement, message verification, supply accounting

Global supply cannot increase through movement

Access/Governance

Role authority, parameters, upgrades

Separate admins, proposal, approval, timelock

EmergencyController.sol

Pause Swap, Borrow, or Bridge separately

Contain damage without seizing user assets

22. Lending Position Lifecycle

State

Allowed action

Validation

Failure result

Deposit

Deposit LQC

Token, vault, price feed

Reject deposit

Borrow

Borrow approved stable asset

LTV, HF, cap, liquidity

Reject loan

Maintain

Add collateral, partial repay

Oracle, interest, HF

Risk alert

Additional borrow

Increase after opt-in

HF, value increase, depth

Stop additional loan

Repay

Repay principal and fee

Amount, fee, burn policy

Debt remains

Withdraw

Recover remaining collateral

Post-withdrawal HF

Reject withdrawal

Liquidate

Dispose part of collateral

HF threshold, Close Factor

Pause or auction review

Interest is not quoted until the model is finalized. Variable/fixed rate, utilization curve, reserve factor, and delinquency concepts are determined with the borrow asset and legal structure.

23. Health Factor and Liquid ation Examp le

Health Factor = (Collateral Value x Liquidation Threshold) / Total Debt

Scenario

Collateral

Threshold

Debt

HF

State

Initial

10,000

60%

4,500

1.333

Healthy

Collateral -15%

8,500

60%

4,500

1.133

Caution

Collateral -25%

7,500

60%

4,500

1.000

Liquidatable

After 50% partial liquidation

7,500

60%

2,250

2.000

Restored

Current technical baseline: maximum LTV 45%, liquidation threshold 60%, Close Factor 50%, and base penalty 5%. Governance may adjust the penalty up to 10% during severe volatility or liquidity deterioration. Liquidation aims to restore HF above 1 while limiting over-liquidation.

24. Oracle Failure Scenario s

State

Entry condition

New loan

Repay

Liquidation

Recovery

NORMAL

Fresh feed, deviation $\leq 5\%$

Allow

Allow

HF rule

Continuous

WARNING

Deviation >5% or delayed update

Reduce cap

Allow

Conservative price

Three normal checks

BORROW_PAUSED

Stale >5 min or sequencer failure

Stop

Allow

Hold/conservative

15-minute grace

EMERGENCY

Multi-feed or chain failure

Stop

Prioritize

May stop

Guardian + multisig

RECOVERY

Feed restored under observation

Stop

Allow

Limited

Five normal checks

The safety priority is to avoid new risk and preserve debt reduction and repayment routes.

25. Liquidity Stress Testing

Test

Assumption

Observation

Example pass condition

Price shock

LQC falls 20-50% in one hour

HF distribution and liquidations

Zero or bounded bad debt

Depth loss

50% of main-pool liquidity withdrawn

Liquidation slippage

Within configured limit

Oracle delay

Price feed stops

Pause time and erroneous loans

No new risk

Stable depeg

USDT/USDC deviates 5-15%

Debt/reserve impact

Asset caps function

Liquidation cascade

Large positions liquidate together

Price impact and failure rate

Close Factor/auction works

Bridge halt

Cross-chain movement unavailable

Chain-level repayment liquidity

Independent operation possible

Stress results determine LTV, Borrow Cap, Debt Ceiling, and liquidation bonus. TVL and loan caps expand only after audit, depth, and operating-data gates pass.

26. Fee and Burn Accounting

Design choice

Benefit

Main risk

Required disclosure

Full payment in LQC

Direct utility

User friction and volatility

Conversion price and grace period

Stable payment followed by LQC purchase

Usability linked to demand

Market impact and execution authority

Purchase limit and route

Partial burn and partial Treasury

Supply reduction plus security funding

Discretionary allocation

Ratio, address, frequency

Full Treasury accumulation

Stronger early safety funding

Misalignment with burn expectations

Use policy and multisig

All burns and financial movements must be verifiable by address, transaction, and periodic report.

Burn may affect circulating supply but does not guarantee token price appreciation.

27. Allocation and Vesting Disclosure Framework

The allocation table in Section 11 is the confirmed design. LQC will reconcile the 1B cap, 150M TGE circulation, wallet-level amounts, monthly circulation forecast, early-release rules, reallocation rights, and approval procedures with on-chain evidence.

28. Governance Transition

Stage

Authority

Scope

Control

Initial deployment

3-of-5 Protocol Multisig

Cap, allowlist, upgrade proposals

48-hour timelock

Risk operations

2-of-3 Risk Multisig

Reduce limits, pause borrowing

Expansion requires timelock

Mature operations

Governance + Security Council

Policy-range changes

Proposal, vote, execution delay

Emergency

Emergency Guardian

Module-level pause only

No transfer or mint authority

No single party may independently control token supply, Treasury, Oracle, upgrades, and Bridge.
Timelock exceptions are limited to emergency pause, not fund movement.

29. Security Threat Model

Attack surface

Representative threat

Prevention

Detection/response

Router

Malicious pool, reentrancy, sandwich

Allowlist, CEI, minOut

output $\geq$ minOut; isolate route

Vault

Accounting error or theft

Least privilege, SafeERC20

assets $\geq$ user claims

Lending

Oracle manipulation, bad debt

Multi-source, caps, HF

borrow $\leq$ limit; immediate pause

Bridge

Forgery, replay, duplicate mint

Domain, nonce, rate limit

one-time nonce; global supply conserved

Treasury

Key theft, arbitrary spending

3-of-5 multisig, 48h timelock

Reject outside policy limits

Upgrade

Malicious implementation, storage collision

Code hash, audit, delay

Preserve authority, balances, supply

Audit scope includes code, deployment scripts, administrative authority, oracle configuration, supply invariants, and economic attacks.

30. Audit, Deployment, and Incident Checklist

Stage

Mandatory check

Evidence

Pre-deployment

Code freeze; unit, fuzz, invariant tests

Zero critical; core branch coverage target $\geq 90\%$

Audit

Independent audit, remediation, retest

Zero Critical/High; disclose Medium

Initial deployment

Low TVL, trade, and loan caps

Publish authority, addresses, parameters

Observation

Reconcile Router, Oracle, accounting

Execution target $\geq 99\%$; zero accounting mismatch

Expansion

Pass KPIs and stress tests

Zero bad debt; P95 quote target $< 1s$

Incident

Module pause and recovery exercise

Detection target < 5 min; publish RCA

Facts and assumptions are separated in incident notices. After recovery, LQC intends to publish root-cause analysis and preventive actions.

31. SDK, API, and Developer Integration

Interface

Main function

Security requirement

Operating metric

Quote API

quoteExactIn, quoteExactOut

Validate chain, token, deadline

P95 <1s; error rate

Execution SDK

buildSwap, executeSwap

minOut, deadline, permit by default

Execution target >=99%

Pool Registry API

getPools, getPoolStatus

Canonical address, version, signed state

Sync and freshness

Lending SDK

deposit, borrow, repay, liquidate

Display HF, oracle state, caps

Zero erroneous loan

Monitoring Feed

events, alerts, moduleState

Replay and missing-event validation

Zero missing event

SDK defaults prioritize safety over convenience and avoid abstractions that hide route and risk information.

32. Exchange Due-diligence Materials

Application, whitepaper, website, explorer, and tokenomics values must agree. Nonexistent links or undeployed contracts are marked N/A or undeployed.

Item

Required evidence

Current status

Project

Official website, whitepaper, GitHub

Verify consistency

Official channels

Domain and account ownership

Addresses to finalize

Contract

Explorer-verified source and deployment tx

Submit after deployment

Tokenomics

Wallet schedule and multisig addresses

Evidence pending

Team/entity

Formation documents, KYC, credentials

KYC package required

Security

Audit report, fixes, tests

Mark incomplete until complete

Market

Listing/CMC/CG links or N/A; MM plan

Submit verified values only

33. Market Integrity and Post- listing Operations

Listing begins ongoing obligations for liquidity, price integrity, security, on-chain maintenance, and disclosure. Wash trading, insider trading, manipulation, and misleading announcements are prohibited.

Domain

Monitoring

Disclosure/response

Liquidity

Depth, two-sided quotes, withdrawal, volume, spread

Adjust funding and caps if insufficient

Price integrity

CEX/DEX divergence, spikes, concentration

Investigate and cooperate with venues

Security

Contract, Bridge, Oracle, keys, failed tx, pauses

Notify, isolate, publish RCA

Operations

Website, social channels, on-chain uptime, development

Announce interruption and recovery

Supply

totalSupply, upgrade, scheduled unlocks

Notice material changes at least 15 days ahead

Market conduct

Related wallets and MM activity

Internal controls and investigation support

Legal/team

Compliance, solvency, sanctions, key personnel

Promptly disclose material facts

34. Required Completion Before an Exchange Submission

Gate

Required result

Current assessment

1

Confirm 150M circulation, allocation, vesting, wallet-level unlock evidence

Design confirmed; evidence pending

2

Deployed contract, verified source, ownership and administration

Blocked / not provided

3

Independent audit; resolve Critical/High; disclose known issues

Blocked / not provided

4

Entity, founder, UBO, team KYC, official contacts

Documentation package required

5

Initial liquidity/MM funding, quoting, withdrawal, conflict policy

Finalization required

6

Website, X, Telegram, GitHub, CMC/CG, listing-link consistency

Verification required

7

Legal opinion, restricted regions, AML/KYC, token classification

Review required

8

Do not claim confirmed listing before formal approval

Effective immediately

The official LQC whitepaper clearly distinguishes objectives from achievements, discussions from contracts, and design values from final deployed values.

Due-diligence content references standard exchange application requirements and market-integrity principles. It does not imply approval, affiliation, endorsement, or a guaranteed listing by any exchange.

END OF WHITEPAPER

Router 2.0, LQC Flow DEX and BSC Testnet Readiness

Version 4.6 - English Working Draft - September 2026

This update forms part of the LQC Whitepaper and records repository-verified implementation progress completed after the August 2026 working draft. It does not represent an audit, mainnet launch, production integration, exchange listing, or guarantee of future performance.

Update classification

Classification	Meaning in this update
Implemented	Code and automated tests are present in the public repository.
Testnet-oriented	Designed for controlled BSC testnet deployment and validation.
Pending	Requires production addresses, parameters, audit, governance approval, monitoring, and legal review.

1. Development snapshot

Component	Repository status	Current scope
LQC Flow AMM	Implemented	Factory, pair, LP accounting, token/BNB liquidity, exact-input and exact-output swaps, multi-hop paths.
Router 2.0	Implemented testnet MVP	Adapter registry, quote isolation, best-route selection, split optimization, atomic execution.
PancakeSwap V2	Adapter implemented	Reviewed-address registration and protected exact-input execution.
PancakeSwap V3	Quote and execution adapters implemented	Validated packed paths, reviewed fee tiers and pools, maximum three hops.
Native BNB routing	Implemented	Wrap/unwrap around protected token execution without retained user balances.
Risk and governance controls	Implemented testnet foundation	Timelock, emergency disable, allowlists, transaction and UTC-day caps.
Production deployment	Pending	Audit, final multisig, oracle feeds, limits, monitoring, legal review and capped pilot.

2. Protocol-neutral routing architecture

Router 2.0 separates discovery, quoting, optimization, execution, risk limits and emergency authority. New EVM DEXs can be integrated through reviewed protocol-specific adapters without replacing the shared router stack.

Contract	Implemented responsibility
LQCDexRegistry	Stores approved adapters, route status and deterministic priority.
LQCQuoteRouter	Compares enabled adapters; isolates failed quote calls; applies priority tie-breaking.
LQCSplitOptimizer	Builds capped candidate allocations and exact output totals across one to four routes.
LQCAutoRouter	Selects single or split execution and derives route-level minimum outputs from user slippage tolerance.
LQCExecutionRouter	Executes exact-input token swaps with deadline, minimum-output, recipient-balance verification, temporary exact approvals and reentrancy protection.
LQCGasCostOracle	Converts estimated BNB gas into output-token units after freshness and deviation checks.
LQCNativeRouter	Wraps and unwraps native BNB without retaining user balances.

2.1 Route selection logic

Enabled execution adapters are quoted independently. Failed DEX quotes are excluded instead of reverting the entire comparison. The router compares expected output after caller-supplied gas cost, applies deterministic registry priority to ties, and executes only through active adapters.

For optimized execution, the system may select one route or an atomic split across two to four routes. User slippage tolerance is converted into per-route minimum-output protection. Tolerance above the configured 20% ceiling is rejected. If any later split leg fails, the whole transaction reverts and restores balances and pool state.

2.2 Adapter admission

Protocol neutrality does not mean automatic compatibility. Each DEX requires a protocol-specific quote adapter, execution adapter, path validation, tests, address review and security review before registry activation. Non-EVM connectivity remains a later cross-chain layer.

3. Implemented controls and trust boundaries

Control	Implemented behavior	Boundary
DEX activation	Only enabled registry adapters may quote or execute.	Governance-controlled expansion.
Minimum output	Transactions revert when protected output is not met.	Depends on user tolerance and quote integrity.
Deadline	Expired execution requests are rejected.	Does not remove market volatility.
Approvals	Exact temporary approvals are cleared after execution.	Token-specific edge cases remain subject to review.
Atomic split	Any failed leg reverts the complete optimized swap.	Gas cost may rise with route count.
Timelock	Structural registry and risk expansion is delayed.	Production target policy remains 48 hours; testnet bootstrap may use one hour.
Emergency controller	Guardians may disable one DEX or pause new swaps immediately.	Guardians cannot resume, re-enable, move user funds, mint, or expand limits.
Risk registry	Token allowlists and per-DEX, per-transaction and UTC-day caps.	Risk multisig may reduce, not expand, limits.

3.1 Gas-aware comparison

The gas-cost oracle validates primary and secondary price freshness and permitted deviation before converting estimated BNB gas into the output-token denomination. This allows comparison by estimated net output rather than gross quote alone.

Production feed addresses, heartbeat windows, deviation thresholds and fallback policy are not finalized. Gas-adjusted selection is an engineering capability, not a guarantee of best realized execution.

3.2 Custody principle

The routing contracts are designed not to retain residual user balances after execution. Native BNB handling is limited to controlled wrapping and unwrapping around the protected token route.

4. Verification completed and gates remaining

The automated suite covers AMM behavior, Router 2.0 logic and deployment wiring. Repository tests include deterministic stateful reserve invariants, capped split invariants, atomic split rollback, route selection, slippage rejection, exact approvals, quote-failure isolation and testnet bootstrap validation.

Verification area	Evidence represented in repository tests
AMM invariants	Reserve accounting and non-decreasing constant-product checks across deterministic state changes.
Split invariants	Exact allocation/output totals and one-to-four-route caps.
Atomic rollback	A later adapter failure restores user balances, pool reserves and zero router/adapter custody.
BSC testnet bootstrap	Pool creation, Router 2.0 quoting, capped smoke swap and minimum-output rejection.
Deployment validation	Chain ID 97 safety, deployed bytecode, Router-to-Factory relationships, registry order, adapter status, module linkage and risk ceilings.

4.1 Deployment status

Deployment tooling targets BSC testnet (chain ID 97). It can deploy test tokens, LQC Flow AMM components and Router 2.0, register the LQC Flow adapter, and optionally register reviewed PancakeSwap V2 or V3 addresses supplied at deployment time.

No production-readiness claim

The implementation remains an unaudited testnet MVP. Fee-on-transfer token support, permit signatures, protocol-fee accounting, LQC fee conversion and burning, production oracle feeds and audited production integrations are intentionally deferred.

Required before mainnet use

Independent audits; invariant and fuzz expansion; economic simulation; final multisig and timelock assignment; production oracle configuration; token, pool and fee-tier allowlists; monitoring and incident procedures; capped-liquidity pilot; and legal and regulatory review.

5. Impact on the LQC roadmap

Roadmap layer	September 2026 interpretation
Foundation	Router, registry, AMM, access and testnet deployment foundations are implemented in repository code.
Router MVP	Multi-adapter quotes, gas-aware comparison, split optimization and protected execution have reached unaudited testnet MVP status.
Controlled integration	LQC Flow plus PancakeSwap V2/V3 adapter paths are implemented; production addresses and live activation remain gated.
Lending	Whitepaper design remains future scope; production lending contracts, oracle configuration and economic validation are not claimed by this update.
Cross-chain	Remains phased future scope. Non-EVM liquidity is not assumed compatible with EVM adapters.
Perpetual trading	Remains a planned module subject to separate liquidity, oracle, margin, liquidation, audit and legal validation.

5.1 Token utility boundary

Router and AMM development does not by itself activate LQC fee conversion, lending-fee burning, staking rewards, governance rights or production Treasury flows. Those utilities remain subject to contract completion, audit, parameter approval and on-chain disclosure.

5.2 Tokenomics status

This technical update does not change token supply, TGE circulation, allocations, vesting or emission schedules. Any final tokenomics values must be approved consistently across the whitepaper, website, contracts, Treasury wallets and public disclosures before launch.

5.3 Team and public documentation

The public project site now provides individual team profile pages and identifies project leadership, engineering, marketing and advisory roles. Team biographies are informational and do not replace technical verification, governance controls or legal disclosures.

Development conclusion

LQC has progressed from router architecture into a broader, testable EVM routing stack with native AMM integration, external DEX adapters, gas-aware selection, atomic split execution and staged risk controls. The next milestone is not unrestricted production deployment; it is controlled BSC testnet validation under final governance, oracle, monitoring and audit preparation.